

MA2003B

Application of Multivariate Methods in Data Science

CIP: 270501 Statistics, General.

CL-L-A-U-CA-ID-AS-AI-CT-HT-S-UDC: 12-0-4-10.7-4-60-20-112-32-192-5-4

Discipline:

Mathematics

School:

Engineering and Sciences

Academic Department:

Sciences

Programs:

5 IDM19

Modules

Prerequisites:

MA1036 and TC2004B

Equivalences:

MA2009B

Course intention within the general study plan context:

This intermediate mathematics course focuses on the analysis of multidimensional large databases. In the process, students identify the characteristics of different problems that can be addressed using multivariate analysis statistical techniques. Prior knowledge is required of probability and statistics, probability functions, normal distribution, and concepts related to inferential statistics: hypothesis testing and confidence intervals. Learning outcome: Students analyze a multidimensional phenomenon or process, in which they will apply one or more of the multivariate statistical techniques learned in the course.

Course objective:

Upon completion of this educational unit, students will be able to:

- Build structured, analyzable information-based databases based in an organization.
- Extract relevant information from large structured databases using statistical methods and appropriate technological tools.
- Formulate the mathematical models most frequently used in large-scale optimization processes.
- Analyze the world as a set of interconnected systems being aware of the need to include an integrated vision in understanding and solving field-related issues.

- Solve problems from diverse environments of life, with ethical awareness, responsibility, and fairness.

Course topics and subtopics:

1. Regression analysis.

1.1 Simple linear regression.

1.2 ANOVA and confidence intervals for predictions.

1.3 Residual analysis and condition of normality.

1.4 Forward selection.

1.5 Backward selection.

1.6 Non-linear regression.

1.7 Multiple linear regression.

1.8 Aberrant data and heteroskedasticity problems.

2.- Multivariate analysis.

2.1 Multivariate distributions.

2.2 Mean vectors and variance and covariance matrices.

2.3 Correlations and correlation matrices.

2.4 Multivariate normal probability density function.

2.5 Lost, null, incorrect values and detection of discrepancies.

2.6 Multivariate aberrant data.

2.7 Sample correlations, Fisher and Ruben intervals.

2.8 Multivariate descriptive analytics and visualization.

3. Principal component analysis (PCA).

3.1 Cases where PCA is used.

3.2 Geometrical description and classification of major components.

3.3 Estimation of major components.

3.4 Determination of the appropriate number of major components.

3.5 Coding and commercial programs.

4. Factor analysis.

4.1 Objectives of factor analysis.

4.2 Factor analysis equations.

4.3 Choice of the appropriate number of factors.

4.4 Factor rotation.

4.5 Oblique rotation method.

4.6 Coding and commercial programs.

5. Discriminant analysis.

5.1 Discrimination for two normal multivariate populations.

5.2 Function of costs and a priori probabilities.

5.3 Basic discrimination.

5.4 Stepwise selection.

5.5 Canonical discriminant functions.

5.6 Coding and commercial programs.

6.- Analysis by conglomerates .

6.1 Similarity and dissimilarity measures.

6.2 Graphical methods (dispersion, main components, Andrews) .

6.3 Non-hierarchical grouping methods.

6.4 Hierarchical grouping.

6.5 Nearest neighbor method.

6.6 Coding and commercial programs .

7.- Multivariate regression.

7.1 Logistic regression model.

7.2 Inferences for variances and covariances matrices.

7.3 Inferences for a vector of means.

7.4 MANOVA.

7.5 Canonical correlation analysis.

7.6 Analysis by factors and regression.

7.7 Programming and commercial systems.

Specific learning objectives by topic:

1. Perform regression analysis.2. Apply multivariate analysis.3. Implement principal component analysis.4. Apply discriminant analysis.5. Implement clustering methods.6. Build multivariate regression models.

Suggested methodologies and learning techniques:

Learning activities led by an academic (Supervised Learning):

1. Review and conceptual analysis conducted by the teacher through learning modules in which they address issues related to multivariate statistical modeling from data.
2. Discussions and resolution of situations related to the challenge faced in the segment.
3. Mentoring and supervision of the field work carried out by the student when facing the challenge in a real environment.
4. Individual and group counseling sessions aimed at supporting the challenge resolution process.

Independent learning activities (Individual Learning):

1. Individual and team research on the fundamentals of regression analysis and multivariate statistical methods.
2. Solution of exercises, problems and cases, both individually and collaboratively, to develop the ability to carry out multivariate statistical analysis procedures to classify, group or model from a series of data.
3. Field work in the context of the real environment in which it faces the challenge related to the appropriate management that goes from the acquisition, storage and transformation of data focused on decision making.
4. Generation of evidence of the development of skills associated with the challenge. Deliverable: a complete report of the solution to the real problem together with the justification of the appropriate statistical method that was chosen.

Teaching and learning technique:

Not Specified

Estimated timing per topic:

Suggested evaluation policies:

Student learning will be evaluated through procedures and criteria to assess the outcome of the learning process in students' performance in the modules, and through performance or product evidence to observe the development of competencies. The evaluation procedures and respective weighted criteria are as follows:

50% --- Participation in and results of activities, assignments, cases and learning module exams, evaluating theoretical and practical knowledge related to the use of regression models and multivariate analysis for data science.

50% --- Performance in the challenge process and outcomes, considering evidence that shows the level of mastery in the following competencies.

Suggested Bibliography:

TEXT BOOKS:

* Johnson, R.A.; Wichern D.W , Applied Multivariate Statistical Analysis. , 6a., Pearson , 2013, Inglés , 9781292024943

BOOKS FOR CONSULTATION:

* Anderson T. W, An introduction to multivariate statistical analysis, 3rd , Hoboken, NJ : John Wiley, 2003, Inglés , 0471360910

* Wickham H., R for Data Science: Import, Tidy, Transform, Visualize, and Model Data, 1a. , Canadá : O'Reilly , 2016, Inglés , 978-1491910399

Support material:

Academic credentials required to teach the course:

(270501)Master Degree in Statistics ; (270101)Master Degree in Mathematics ; (270301)Master Degree in Applied Mathematics ;

(400101)Master Degree in Physical Sciences ; (400801)Master Degree in Physics ; (270501)Doctoral Degree in Statistics ;

(270101)Doctoral Degree in Mathematics ; (270301)Doctoral Degree in Applied Mathematics ; (400101)Doctoral Degree in Physical Sciences ; (400801)Doctoral Degree in Physics

CIP: 270501, 270101, 270301, 400101, 400801